```
"aggregations" : {
  "extended_stats" : {
    "count" : 6,
    "min" : 999.0,
    "max" : 1800.0,
    "avg" : 1299.6666666666667,
    "sum" : 7798.0,
    "sum_of_squares" : 1.0655002E7,
    "variance" : 86700.22222222232,
    "variance_population" : 86700.22222222232,
    "variance_sampling" : 104040.2666666668,
    "std_deviation" : 294.4490146395846,
    "std_deviation_population" : 294.4490146395846,
    "std_deviation_sampling" : 322.5527347065388,
    "std_deviation_bounds" : {
      "upper" : 1888.5646959458359,
      "lower" : 710.7686373874975,
      "upper_population" : 1888.5646959458359,
      "lower_population" : 710.7686373874975,
      "upper_sampling" : 1944.7721360797443,
      "lower_sampling" : 654.5611972535892
    }
  }
}
```


Figure 13.1 The extended statistics on the `price_gbp` field

13.2.8 The cardinality metric

The `cardinality` metric returns unique values for the given set of documents. A single-value metric fetches occurrences of distinct values from our data. For example, the query in the next listing retrieves the number of unique TV brands in our `tv_sales` index.

Listing 13.9 Fetching unique TV brands

```
GET tv_sales/_search
{
  "size": 0,
  "aggs": {
    "unique_tvs": {
      "cardinality": {
        "field": "brand.keyword"
      }
    }
  }
}
```

The cardinality metric fetches unique values.

Applies cardinality to the brand.keyword field

Because we have four unique brands (Samsung, LG, Philips, and Panasonic), the query result has `4` in the `unique_tvs` aggregation:

```
"aggregations" : {
  "unique_tvs" : {
    "value" : 4
  }
}
```

Data is distributed in Elasticsearch, so trying to fetch exact counts of cardinality may lead to performance problems. To fetch an exact number, data must be retrieved and